Jiale Shao

Robotics & Mechatronics | Mechanism Design | Embedded Prototyping

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Seeking: 6-month placement, Apr – Sep 2027 · robotics hardware / mechatronics / embedded · open to relocate

MEng Design Engineering student at Imperial College London who enjoys turning robot ideas into working hardware. I built the balance control for a self-balancing robot, led the mechanism design of a robotic planter, and saw an aero package go from my CAD to real carbon parts at Chery Racing Development. Portfolio with photos and videos: jerrold-jiale0623.github.io

EDUCATION

Imperial College London — MEng Design Engineering 2024 – 2028 (Expected)

Modules: Robotics, Electronics & Circuit Design, Computer Science, Solid Mechanics, Finite Element Analysis, Fluid & Thermal Dynamics, Human-Centred Design Engineering

Awards: **WRO (World Robot Olympiad) Provincial Champion** (2023); **School Principal's First-Class Scholarship** (Top 3%, 2021–2024)

ROBOTICS & MECHATRONICS PROJECTS

Self-balancing Segway — DE2 Electronics coursework, pair project 2026

- Built up a two-wheeled robot (MicroPython, motion sensor, two encoder motors) through five open challenges, from tilting a board to steer it, to dancing in time with music, to **balancing upright on its own**.
- Wrote **PID speed control for each wheel** from encoder counts, so the wheels kept their speed even when I pushed against them by hand.
- Wrote the **balance controller**: combined accelerometer and gyro readings into one steady tilt angle, then tuned the gains live with a knob, finding the stable range one term at a time until the robot stood up by itself.

Hachi-Pot Bionic Smart Planter — Team project, mechanical lead 2026

- **Led CAD, structure and mechanism** for an ESP32 robot planter that senses light, soil moisture and touch, and shows its mood through **changing mechanical faces**.
- Designed in Fusion 360 the moving parts that **lift the plant, switch the face and drive the pot around**, packed inside a compact body.
- Our first prototype left no room for the electronics. My teammate and I chose to **fix the hardware rather than work around it**: I redrew all 8 snap-fit, glue-free printed layers, and Rev B assembled cleanly and worked at the demo. **Lesson:** plan space for wiring from the first sketch.

Stackin' Beats — CTO, Imperial team project (4 members) 2026

- Owned the circuit, cam mechanism and working prototype of a tactile music kit: players stack coded bricks to compose a melody, and a motor-driven camshaft turns it into a moving wave.

PROFESSIONAL EXPERIENCE

Chery Racing Development, Chery Automobile — Aerodynamics Intern Jul – Sep 2026

Global Technology Innovation Center

- **Designed in CAD the next-generation aero package** for a 330 electric performance sedan, including an **adjustable rear-wing angle mechanism** and swan-neck wing mounts, against downforce and drag targets I set.
- Worked with a CFD colleague over several design rounds that **more than tripled the car's downforce**, then followed the first carbon kit through **manufacturing at the factory**, checking that what was built matched the design.

ENGINEERING ANALYSIS PROJECTS

- **Prosthetic running blade:** tripled its lowest natural frequency so it no longer resonates while running, and checked strength and fatigue in Ansys.
- **Electric GT car aero:** reshaped the body in CFD to **cut drag by about a third**, then tested a 3D-printed model in a wind tunnel.

TECHNICAL SKILLS

Electronics & Control: PID control, sensor fusion; C++, Python/MicroPython, MATLAB; Arduino, ESP32; DC / servo / stepper motors and drivers; sensors, bench testing

CAD / CAE: Fusion 360, SolidWorks, Ansys Mechanical/Workbench, Ansys Fluent

Prototyping & DFMA: FDM/SLA 3D printing, laser cutting, tolerancing, snap-fits, design for assembly

Languages: Chinese (native), English (fluent)